

D844

General Chemistry I Lab

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A student at one of the lab's 24 benches wonders whether water temperature affects how fast an antacid tablet dissolves. The tablets come in a blue box. Which is a testable hypothesis?

- A. Antacid tablets are interesting to study.
- B. Tablets from a blue box dissolve better.
- C. If water temperature increases, the tablet will dissolve faster, because particles collide more often.
- D. Temperature always controls everything about dissolving.

Target word _____	The one fact that decides _____	Answer ___	[] sure [] unsure [] guess
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2.

In the antacid experiment, the student uses water at 10 °C, 25 °C, and 40 °C, times how long each tablet takes to dissolve, and uses 200 mL of water and the same brand every time. The lab room just got a new fume hood. What is the dependent variable?

- A. The water temperature
- B. The time for the tablet to dissolve
- C. The new fume hood
- D. The 200 mL volume of water

Target word _____	The one fact that decides _____	Answer ___	[] sure [] unsure [] guess
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3. [TRANSFER — reversed stem / different variable]

A student adds 0 g, 5 g, and 10 g of salt to 200 mL samples of water and measures each boiling point. What is the independent variable?

- A. The boiling point
- B. The 200 mL volume
- C. The mass of salt added
- D. There is none, since all variables are always held constant.

Target word _____	The one fact that decides _____	Answer ___	[] sure [] unsure [] guess
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4.

A student tests whether a fertilizer increases the growth of bean plants, using 20 plants in a greenhouse built in 1998. Which is the best control group?

- A. No control is needed if the plants grow tall.
- B. Identical plants with the same soil, water, and light, but no fertilizer.
- C. Plants with no fertilizer in a different greenhouse with different light.
- D. Plants given double the fertilizer.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

The accepted density of aluminum is 2.70 g/cm^3 . In a 50-minute lab session, a student's four trials give 2.91, 2.92, 2.90, and 2.91 g/cm^3 . How are the results best described?

- A. Accurate and precise
- B. Neither accurate nor precise
- C. Accurate but not precise
- D. Precise but not accurate

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

The accepted value is 10.0. Which data set is accurate but not precise?

- A. 12.1, 12.0, 12.1, 12.2
- B. 9.0, 11.2, 10.1, 9.7
- C. 10.0, 10.1, 9.9, 10.0
- D. 7.0, 13.5, 8.2, 14.9

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

In a class of 28 students, one student measures the density of aluminum as 2.83 g/cm^3 . The accepted value is 2.70 g/cm^3 . What is the percent error?

- A. About 104.8%
- B. 0.13
- C. About 4.6%
- D. About 4.8%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

A measurement has a 5.0% error compared with an accepted value of 40.0 g, and the measured value was higher than accepted. What was the measured value?

- A.** 42.0 g
- B.** 45.0 g
- C.** 38.0 g
- D.** 2.0 g

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

At a school lab with 30 microscopes, a student measures a block face as 2.35 cm by 1.2 cm. How should the area be reported?

- A.** 2.8 cm²
- B.** 3 cm²
- C.** 2.82 cm²
- D.** 2.820 cm²

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

A student adds masses of 12.11 g, 0.3 g, and 1.052 g. How should the total be reported?

- A.** 10 g
- B.** 13.46 g
- C.** 13.462 g
- D.** 13.5 g

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

In a titration, one of five trials is far off, and the student's notebook shows they overshoot the endpoint on that trial. The instructor has 12 years of teaching experience. What should the student do?

- A.** Keep the trial in the record with the noted error, and exclude it from the average with that reason stated (or repeat the trial).
- B.** Always average every value, whatever happened.
- C.** Erase the trial from the notebook.
- D.** Change the value to match the others.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A student's lab report, due Friday on the course site, includes this sentence: "The rate likely doubled at 40 °C because more particle collisions had enough energy to react." Which section should it go in?

- A.** Introduction, since explanations always go there
- B.** Results
- C.** Discussion
- D.** Methods

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A student ~~at one of the lab's 24 benches~~ wonders whether water temperature affects how fast an antacid tablet dissolves. ~~The tablets come in a blue box.~~ Which is a testable hypothesis?

Target word: testable hypothesis **Decides:** A falsifiable prediction linking temperature to dissolving time.

A	Dropped element	No variables or prediction.
B	Camouflage	Box color is not the variable in question.
C	Correct	Names the variables, predicts a direction, and gives a reason.
D	Absolute language	Sweeping and not a specific, testable prediction.

2. Answer B Concept 2

In the antacid experiment, the student uses water at 10 °C, 25 °C, and 40 °C, times how long each tablet takes to dissolve, and uses 200 mL of water and the same brand every time. ~~The lab room just got a new fume hood.~~ What is the dependent variable?

Target word: dependent variable **Decides:** What is measured.

A	Inverted operation	That is set by the student — the independent variable.
B	Correct	It is measured in response to temperature.
C	Camouflage	Not part of the experiment design.
D	Confusion set	Held constant; a controlled variable.

3. Answer C Concept 2 · transfer

A student adds 0 g, 5 g, and 10 g of salt to 200 mL samples of water and measures each boiling point. What is the independent variable?

Target word: reverse: identify the independent variable **Decides:** What the student deliberately changes.

A	Inverted operation	That is measured — the dependent variable.
B	Confusion set	Held constant.
C	Correct	It is set by the experimenter.
D	Absolute language	An experiment must change one variable.

4. Answer B Concept 3

A student tests whether a fertilizer increases the growth of bean plants, using 20 plants ~~in a greenhouse built in 1998.~~ Which is the best control group?

Target word: proper control **Decides:** Only the fertilizer should differ.

A	Absolute language	Without a control, growth cannot be tied to the fertilizer.
B	Correct	Isolates the fertilizer's effect.
C	Dropped element	Drops the matching conditions, so light confounds the result.
D	Adjacent category	Another treatment level, not a control.

5. Answer D Concept 4

The accepted density of aluminum is 2.70 g/cm^3 . ~~In a 50-minute lab session,~~ a student's four trials give 2.91, 2.92, 2.90, and 2.91 g/cm^3 . How are the results best described?

Target word: accuracy and precision **Decides:** Tight cluster about 0.21 above the accepted value.

A	Confusion set	Tight clustering is not closeness to the accepted value.
B	Adjacent category	The spread is only 0.02, so they are precise.
C	Inverted operation	Reverses the two terms.
D	Correct	Close to each other but far from 2.70.

6. Answer B Concept 4 • transfer

The accepted value is 10.0. Which data set is accurate but not precise?

Target word: reverse: pick the matching data **Decides:** Average near 10.0 with wide spread.

A	Inverted operation	Precise but not accurate.
B	Correct	Mean 10.0, but the values spread widely.
C	Confusion set	Accurate and precise.
D	Adjacent category	Mean 10.9 with wide spread — neither.

7. Answer D Concept 5

~~In a class of 28 students,~~ one student measures the density of aluminum as 2.83 g/cm^3 . The accepted value is 2.70 g/cm^3 . What is the percent error?

Target word: percent error **Decides:** $0.13 \div 2.70 \times 100$.

A	Adjacent category	Experimental $\div$ accepted $\times 100$, not the error.
B	Dropped element	The raw difference, not a percent.
C	Inverted operation	Divides by the experimental value, 2.83.
D	Correct	$0.13 \div 2.70 = 0.0481$.

8. Answer A Concept 5 • transfer

A measurement has a 5.0% error compared with an accepted value of 40.0 g, and the measured value was higher than accepted. What was the measured value?

Target word: solve for the measured value **Decides:** Difference = $0.050 \times 40.0 = 2.0 \text{ g}$, above accepted.

A	Correct	$40.0 + 2.0$.
B	Wrong stem number	Adds 5 g instead of 5%.
C	Inverted operation	Subtracts; the stem says the measurement was higher.
D	Dropped element	The difference alone.

9. Answer A Concept 6

~~At a school lab with 30 microscopes,~~ a student measures a block face as 2.35 cm by 1.2 cm. How should the area be reported?

Target word: area with correct significant figures **Decides:** 1.2 has 2 significant figures.

A	Correct	2.82 rounds to 2 significant figures.
B	Confusion set	Rounds to 1 significant figure; 1.2 has 2.
C	Dropped element	Keeps a digit the 1.2 cm measurement cannot support.
D	Adjacent category	Adds false precision.

10. Answer D Concept 6 • transfer

A student adds masses of 12.11 g, 0.3 g, and 1.052 g. How should the total be reported?

Target word: addition rule **Decides:** Fewest decimal places is one (0.3 g).

A	Inverted operation	Applies the multiplication rule (fewest significant figures).
B	Adjacent category	Two decimal places; 0.3 g limits it to one.
C	Dropped element	Keeps all calculator digits.
D	Correct	13.462 rounds to one decimal place.

11. Answer A Concept 7

In a titration, one of five trials is far off, and the student's notebook shows they overshot the endpoint on that trial. ~~The instructor has 12 years of teaching experience.~~ What should the student do?

Target word: handling an anomalous trial **Decides:** A documented procedural error.

A	Correct	Transparent, justified exclusion keeps the data honest.
B	Absolute language	A known procedural error justifies exclusion with a stated reason.
C	Dropped element	Removes raw data from the record.
D	Inverted operation	Altering data is fabrication.

12. Answer C Concept 8

~~A student's lab report, due Friday on the course site,~~ includes this sentence: "The rate likely doubled at 40 °C because more particle collisions had enough energy to react." Which section should it go in?

Target word: report section **Decides:** The sentence explains why — interpretation.

A	Absolute language	The Introduction gives background, not interpretation of your data.
B	Confusion set	Results present data without interpreting it.
C	Correct	Interpretation and explanation belong in the Discussion.
D	Adjacent category	Methods describe procedure.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	7	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	7	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	8	
Absolute language	Always, never, guarantee — where judgment applies.	5	
Confusion set	Every option is real; the test is exact role-matching.	6	
Camouflage	Answers a stem fact that was never the question.	2	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Testable hypothesis	A hypothesis is a testable, falsifiable prediction linking a change in one variable to an outcome, ideally with a reason.	Writing an opinion, a vague claim, or a statement that cannot be proven wrong.	"If ... then ... because ..." vs. general statements.
2	Independent, dependent, and controlled variables	The independent variable is deliberately changed, the dependent variable is measured, and controlled variables are held constant.	Swapping independent and dependent, or calling a held-constant factor the dependent variable.	What the experimenter sets vs. what is recorded.
3	Control groups	A control group matches the experimental group in every way except the factor being tested.	Using a control that differs in several ways, or skipping it.	"Does treatment X cause change Y?"
4	Accuracy vs. precision	Accuracy is closeness to the accepted value; precision is closeness of repeated results to each other.	Treating tightly clustered results as accurate.	Repeated trials next to an accepted value.
5	Percent error	Percent error = $ \text{experimental} - \text{accepted} \div \text{accepted} \times 100$.	Dividing by the experimental value or reporting the raw difference.	A measured value and an accepted value.
6	Significant figures	Multiplication and division: keep the fewest significant figures; addition and subtraction: keep the fewest decimal places.	Copying every calculator digit or mixing up the two rules.	Measurements with different precision combined in one calculation.
7	Data integrity	Record all raw data; exclude a result only for a documented, stated reason, and never alter values.	Erasing or adjusting an inconvenient trial.	One trial far from the others.
8	Lab report sections	Methods describe what was done so it can be repeated; Results present data without interpretation; Discussion explains and interprets.	Placing explanations in Results.	A sentence with "because" or "likely" about why a result happened.

Course scope (public description): General Chemistry I Lab focuses on developing foundational skills in scientific investigation in chemistry. It emphasizes the application of the scientific method to answer chemistry questions through hypothesis-driven experimentation. Students will learn to design, execute, and analyze chemistry experiments, ensuring adherence to rigorous scientific protocols and ethical standards. The course also covers essential aspects of scientific communication, including writing clear and structured scientific reports and effectively presenting experimental findings. Throughout the course, students will cultivate critical thinking skills necessary for interpreting data and drawing conclusions.